

Scenario: Management – adults

Last revised in March 2025

From age 18 years onwards.

What initial information and advice should I offer an adult with type 2 diabetes?

Ensure that the person has an individualized care plan, taking into account their age, preferences, co-morbidities, and risk of adverse effects from polypharmacy and their likelihood of benefiting from long-term interventions.

- Offer referral to a structured group education programme such as the DESMOND (Diabetes Education for Self-Management for Ongoing and Newly Diagnosed) programme, for the person and their family/carers.
 - This should be offered at or around the time of diagnosis, with annual reinforcement and review.
 - Advise that the website www.desmond.nhs.uk (<https://www.desmond.nhs.uk/>) may have useful information.
- Ensure that the person and their family/carers know how to contact the specialist diabetes team, if appropriate.
- Provide advice on sources of information and support.
 - Diabetes UK (website available at www.diabetes.org.uk (<https://www.diabetes.org.uk/>)) — a national charity that provides information, support, and advocacy for people with diabetes and their families. It has a confidential helpline (0345 123 2399) and hosts an online community for peer support, as well as a wide range of education and information [patient resources](https://www.diabetes.org.uk/professionals/information-support-for-your-patients/education-information) (<https://www.diabetes.org.uk/professionals/information-support-for-your-patients/education-information>).
 - For detailed information on managing other lifestyle and cultural issues, such as driving, fasting, holiday and travel, insurance, and shift work, see the CKS topic on [Insulin therapy in type 2 diabetes](#) (</topics/insulin-therapy-in-type-2-diabetes/>).
- Provide advice on [lifestyle measures](/topics/diabetes-type-2/management/management-adults/#lifestyle-advice) (</topics/diabetes-type-2/management/management-adults/#lifestyle-advice>), such as diet, exercise, and weight loss, if appropriate.
- Assess for associated anxiety and depression, and manage appropriately. See the CKS topics on [Generalized anxiety disorder](/topics/generalized-anxiety-disorder/) (</topics/generalized-anxiety-disorder/>) and [Depression](/topics/depression/) (</topics/depression/>) for more information.
 - The Diabetes UK patient resource [Diabetes and your emotions](https://www.diabetes.org.uk/guide-to-diabetes/emotions) (<https://www.diabetes.org.uk/guide-to-diabetes/emotions>) may be helpful.
- Advise about sexual health, contraception, and the importance of pre-pregnancy counselling, if appropriate. See the CKS topics on [Contraception – assessment](/topics/contraception-assessment/) (</topics/contraception-assessment/>) and [Pre-conception – advice and management](/topics/pre-conception-advice-management/) (</topics/pre-conception-advice-management/>) for more information.
 - The Diabetes UK patient resource [Planning for a pregnancy when you have diabetes](https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/pregnancy) (<https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/pregnancy>) may be helpful.

- Offer immunization against influenza and pneumococcal infection. See the CKS topics on [Immunizations – seasonal influenza \(/topics/immunizations-seasonal-influenza/\)](#) and [Immunizations – pneumococcal \(/topics/immunizations-pneumococcal/\)](#) for more information.
- Advise about entitlement to free NHS prescriptions if the person uses insulin or antidiabetic medication. See the Diabetes UK section on [Free prescriptions \(England\) \(https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/free-prescriptions\)](#) for more information.
- Advise the person to always wear or carry some form of diabetes identification, such as a:
 - MedicAlert® bracelet, necklace, or watch (available from [www.medicalert.org.uk \(http://www.medicalert.org.uk/\)](#)).
 - Diabetes identity card or wristband (available from [www.diabetes.org.uk \(https://www.diabetes.org.uk/\)](#)).
- Advise that HbA1c should be measured at 3–6 monthly intervals initially until stable on unchanging antidiabetic treatment, and then every 6 months to ensure adequate blood glucose control. See the section on [Treatment targets \(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#) for more information.
- Advise that [screening for complications \(/topics/diabetes-type-2/management/management-adults/#screening-for-managing-complications\)](#) will be arranged at diagnosis and then annually thereafter.

Basis for recommendation

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What are the treatment targets for adults with type 2 d

Offer [lifestyle advice \(/topics/diabetes-type-2/management/management-adults/#lifestyle-advice\)](#) and [drug treatment\(s\) \(/topics/diabetes-type-2/management/management-adults/#antidiabetic-drugs\)](#) to achieve an individualized target HbA1c level, to minimize the risk of long-term [complications \(/topics/diabetes-type-2/background-information/complications/\)](#), depending on the person's age, preferences, risk of adverse effects, and co-morbidities.

- Educate the person about their individual recommended HbA1c target, and encourage measures to achieve and maintain it, where possible.
 - Lifestyle including diet management — 48 mmol/mol (6.5%).
 - Lifestyle including diet combined with a single drug not associated with hypoglycaemia (such as metformin) — 48 mmol/mol (6.5%).
 - Drug treatment associated with hypoglycaemia (such as a sulfonylurea): 53 mmol/mol (7.0%).
 - Consider using the [NICE patient decision aid \(https://www.nice.org.uk/guidance/ng28/resources/patient-decision-aid-2187281197\)](#) to support these decisions.
- Measure HbA1c levels at 3–6 monthly intervals until the HbA1c is stable on unchanging treatment, then at 6-monthly intervals.
- If HbA1c levels are not adequately controlled by a single drug and rise to 58 mmol/mol (7.5%) or higher:
 - Reinforce advice on lifestyle including diet.
 - Assess the person's adherence to antidiabetic drug treatment.
 - Support the person to aim for an HbA1c level of 53 mmol/mol (7.0%).

- Intensify antidiabetic drug treatment, if appropriate. See the section on [Antidiabetic drugs \(/topics/diabetes-type-2/management/management-adults/#antidiabetic-drugs\)](#) for more information.
- **HbA1c targets should be individualized according to** the person's duration of diabetes, co-morbidities, and age, and may be reduced if:
 - There are adverse effects from treatment (such as hypoglycaemia), or treatment is affecting the person's quality of life.
 - Tight blood glucose control and possible hypoglycaemia may be inappropriate for people at risk of falls, those with impaired awareness of hypoglycaemia, and people who drive or operate machinery.
 - The person is unlikely to achieve longer-term risk-reduction benefits, for example if they have significant co-morbidities or a reduced life expectancy.
- **If the person achieves an HbA1c level that is lower than their target** and there are no adverse effects such as hypoglycaemia:
 - Encourage them to maintain it.
 - Be aware of alternative reasons for a low HbA1c level, such as deteriorating renal function or sudden weight loss. See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](#) for more information.
 - Consider stopping any medication that is not effective at review.
- **Do not routinely recommend self-monitoring of blood glucose levels.**
 - Advise the person on the need to self-monitor blood glucose levels if they:
 - Are using insulin therapy. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for more information.
 - Have evidence of hypoglycaemic episodes.
 - Are taking a drug that may increase the risk of hypoglycaemia while driving or operating machinery (such as a sulfonylurea).
 - Are pregnant or planning a pregnancy. See the CKS topic on [Pre-conception - advice and management \(/topics/pre-conception-advice-management/\)](#) for more information.
 - Note: advise that blood glucose levels may increase with acute intercurrent illness or infection. See the section on [Managing intercurrent illness \(/topics/diabetes-type-2/management/management-adults/#managing-intercurrent-illness\)](#) for more information.
 - Provide appropriate advice on driving if the person is self-monitoring blood glucose levels.
 - See the Driver and Vehicle Licensing Agency (DVLA) document [Assessing fitness to drive: a guide for medical professionals \(https://www.gov.uk/government/publications/assessing-fitness-to-drive-a-guide-for-medical-professionals\)](#) for more information.
 - Consider the need for short-term self-monitoring of blood glucose levels:
 - When starting treatment with corticosteroids, *or*
 - To confirm suspected hypoglycaemia ([/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/](#)).
- **If routine self-monitoring of blood glucose levels is indicated**, review the person at least annually to check:
 - Self-monitoring skills, the quality and frequency of testing, and the equipment used. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for information on blood glucose meters, lancets, and strips.
 - The person knows how to interpret blood glucose results, and what action to take if they are too high or too low.

- The impact of self-monitoring on the person's quality of life and the continued benefit to the person.

Basis for recommendation

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Which antidiabetic drugs are available for adults with t

If an adult with type 2 diabetes is hyperglycaemic with [symptoms \(/topics/diabetes-type-2/diagnosis/diagnosis-in-adults/\)](#), consider the need for immediate insulin therapy or treatment with a sulfonylurea depending on specialist advice, and review treatment when blood glucose control has been achieved.

For all other adults, offer individualized antidiabetic treatment depending on the person's HbA1c [treatment target \(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#), age, preferences, comorbidities, contraindications, weight, risks and benefits of treatment, and polypharmacy. Consider the effectiveness of the medications prescribed in terms of metabolic response, cardiovascular and renal protection, safety and tolerability. Consider also the need for monitoring, licensed indications, combination of medications and cost, choosing the drugs with the lower acquisition cost.

- **Offer standard-release metformin as first-line treatment, unless it is contraindicated.**
 - Gradually increase the dose of standard-release [metformin \(/topics/diabetes-type-2/prescribing-information/metformin/\)](#) over several weeks to minimize the risk of adverse effects.
 - If there are intolerable gastrointestinal adverse effects, consider a trial of modified-release metformin.
- **Assess the person's cardiovascular status** and risk to determine whether they:
 - Have chronic heart failure,
 - Have established atherosclerotic cardiovascular disease, *or*
 - Are at high risk of developing cardiovascular disease.
- **Based on this cardiovascular risk assessment** for the person with type 2 diabetes:
 - If they have chronic heart failure or established atherosclerotic cardiovascular disease, offer an SGLT-2 inhibitor with proven cardiovascular benefit in addition to metformin.
 - If they are at high risk of developing cardiovascular disease, consider an SGLT-2 inhibitor with proven cardiovascular benefit in addition to metformin.
- When starting an adult with type 2 diabetes on dual therapy with metformin and an SGLT-2 inhibitor as first-line therapy, introduce the drugs sequentially, starting with metformin and checking tolerability. Start the SGLT-2 inhibitor as soon as metformin tolerability is confirmed.
- **If metformin is contraindicated or not tolerated:**
 - Offer an SGLT-2 inhibitor to people with chronic heart failure or established atherosclerotic cardiovascular disease, and consider offering an SGLT-2 inhibitor to people at high risk of developing cardiovascular disease.
 - For other people, consider one of the following as first-line treatment:
 - A dipeptidyl peptidase-4 inhibitor ([DPP-4 inhibitor \(/topics/diabetes-type-2/prescribing-information/dpp-4-inhibitors/\)](#)).
 - [Pioglitazone \(/topics/diabetes-type-2/prescribing-information/pioglitazone/\)](#).
 - A [sulfonylurea \(/topics/diabetes-type-2/prescribing-information/sulfonylureas/\)](#).

- A sodium-glucose cotransporter-2 inhibitor ([SGLT-2 inhibitor \(/topics/diabetes-type-2/prescribing-information/sglt-2-inhibitors/\)](#)) — this may be considered instead of a DPP-4 inhibitor if a sulfonylurea or pioglitazone is not appropriate.
- **For adults with type 2 diabetes at any stage after they have started first-line treatment:**
 - If they have or develop chronic heart failure or established atherosclerotic cardiovascular disease, offer an SGLT-2 inhibitor with proven cardiovascular benefit in addition to current treatment or replace an existing drug with the SGLT-2 inhibitor.
 - If they are or become at high risk of developing cardiovascular disease, consider adding an SGLT-2 inhibitor with proven cardiovascular benefit to current treatment or replacing an existing drug with the SGLT-2 inhibitor.
 - Take into account the person's current treatment regimen and preferences and make a shared decision about switching treatments or adding an SGLT-2 inhibitor, as appropriate.
- **If monotherapy is ineffective ([\(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#)),** consider adding one of the following:
 - A DPP-4 inhibitor.
 - Pioglitazone.
 - A sulfonylurea.
 - An SGLT-2 inhibitor — this may be considered in people taking metformin if a sulfonylurea is contraindicated or not tolerated, or the person is at significant risk of hypoglycaemia or its consequences.
- **If dual therapy is ineffective ([\(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#)):**
 - For people who can take metformin, consider:
 - Triple therapy by adding a DPP-4 inhibitor, pioglitazone, a sulfonylurea, or an SGLT-2 inhibitor (canagliflozin or empagliflozin). Note: dapagliflozin is recommended only in combination with metformin and a sulfonylurea, *not* pioglitazone, and ertugliflozin only in combination with metformin and a DPP-4 inhibitor, if a sulfonylurea or pioglitazone is not appropriate.
 - Starting insulin-based treatment. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for more information.
 - For people in whom metformin is contraindicated or not tolerated, consider starting insulin-based treatment. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for more information.
- **If triple therapy is ineffective ([\(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#)), not tolerated, or contraindicated:**
 - Consider switching one drug for a glucagon-like peptide-1 receptor agonist ([GLP-1 receptor agonist \(/topics/diabetes-type-2/prescribing-information/glp-1-receptor-agonists/\)](#)) for people with:
 - A body mass index (BMI) of 35 kg/m² or higher (adjust accordingly for people from black, Asian, and other minority ethnic groups) and specific psychological or other medical conditions associated with obesity, *or*
 - A BMI lower than 35 kg/m², and insulin therapy would have significant occupational implications, or weight loss would benefit other significant obesity-related comorbidities.
 - Note: a GLP-1 receptor agonist should only be continued if the person has shown a reduction of at least 11 mmol/mol [1.0%] in HbA1c *and* a weight loss of at least 3% of initial body weight in 6 months.
- **For people on insulin-based treatment, only offer combination therapy with a GLP-1 receptor agonist plus insulin after taking specialist advice, and with ongoing support**

from a consultant-led multidisciplinary team.

- Note: if a GLP-1 receptor agonist is added to insulin or an insulin secretagogue (sulfonylurea or meglitinide), consider reducing the dose of the latter agents to reduce the risk of [hypoglycaemia \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/\)](#). If a DPP-4 inhibitor is being taken, it should be discontinued on starting a GLP-1 receptor agonist, as the combination does not provide additional blood glucose control.

Basis for recommendation

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What lifestyle advice should I give to adults with type 2

Diet

Offer appropriate, individualized, and ongoing nutritional advice from a healthcare professional with specific expertise and competencies in nutrition.

- **Emphasize the importance of a healthy, balanced diet:**
 - Encourage high fibre, low-glycaemic-index sources of carbohydrate (such as fruit, vegetables, wholegrains, and pulses), low-fat dairy products, and oily fish.
 - Advise to control the intake of foods containing saturated and trans-fatty acids, high-sugar drinks, and foods high in salt content.
 - Individualize recommendations for carbohydrate and alcohol intake, and regular meal patterns, to reduce the risk of [hypoglycaemia \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/\)](#), especially in people being treated with insulin or a [sulfonylurea \(/topics/diabetes-type-2/prescribing-information/sulfonylureas/\)](#). See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for more information.
 - Advise that limited substitution of sucrose-containing foods for other carbohydrate in the meal plan is allowed, but care should be taken to avoid excess energy intake.
 - Discourage the use of foods marketed specifically for people with diabetes.
 - The Diabetes UK patient resources [Weight loss and diabetes \(/https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/eating-with-diabetes/whats-your-healthy-weight/lose-weight\)](#) and [What is a healthy, balanced diet for diabetes? \(/https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/eating-with-diabetes/what-is-a-healthy-balanced-diet\)](#) may be helpful.
- **Integrate dietary advice with other aspects of lifestyle modification, such as exercise and weight loss, as appropriate.**
 - If the person is overweight or obese, set an initial body weight loss target of 5–10% (smaller weight losses may still be beneficial). See the CKS topic on [Obesity \(/topics/obesity/\)](#) for further information on weight management.

Exercise and physical activity

- **Provide information on regular exercise and physical activity.**
 - The Diabetes UK patient resource [Diabetes and exercise \(/https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/exercise\)](#) provides helpful information on the type and duration of recommended exercise.
 - Aim to minimize time spent being sedentary.

- Advise that regular exercise may lower blood glucose levels, improve cardiovascular risk, and reduce excess weight (when combined with a [healthy diet](#) ([/topics/diabetes-type-2/management/management-adults/#diet](#))). See the CKS topic on [Obesity](#) ([/topics/obesity/](#)) for more information.
- **Provide information on the following (if appropriate):**
 - The potential interaction of exercise and alcohol.
 - The effect of exercise on blood glucose levels when insulin levels are adequate (risk of [hypoglycaemia](#) ([/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/](#))).
 - The effect of exercise on blood glucose levels when the person is hyperglycaemic (risk of worsening blood glucose control and ketones in the blood).
 - The appropriate adjustments of insulin dosage and/or nutritional intake for exercise and post-exercise periods. See the CKS topic on [Insulin in type 2 diabetes](#) ([/topics/insulin-therapy-in-type-2-diabetes/](#)) for more information.

Alcohol intake

- **Provide information on recommended alcohol limits and the potential impact of alcohol on diabetes and its treatment.**
 - Advise on the recommended alcohol limits. See CKS topic on [Alcohol - problem drinking](#) ([/topics/alcohol-problem-drinking/](#)) for more information.
 - Advise the person to ideally eat a snack that contains carbohydrate before and after drinking alcohol.
 - Educate that alcohol may exacerbate or prolong the hypoglycaemic effect of [antidiabetic drugs](#) ([/topics/diabetes-type-2/management/management-adults/#antidiabetic-drugs](#)).
 - The signs of [hypoglycaemia](#) ([/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/](#)) may become less obvious, and delayed hypoglycaemia may occur up to several hours after alcohol consumption. See the CKS topic on [Insulin therapy in type 2 diabetes](#) ([/topics/insulin-therapy-in-type-2-diabetes/](#)) for more information on the possible presentation and effects of hypoglycaemia.
 - Advise the person to always wear or carry some form of diabetes identification, as reduced awareness of hypoglycaemia may be confused with alcohol intoxication, such as a:
 - MedicAlert® bracelet, necklace, or watch (available from [www.medicalert.org.uk](#) ([http://www.medicalert.org.uk/](#))).
 - Diabetes identity card or wristband (available from [www.diabetes.org.uk](#) ([https://www.diabetes.org.uk/](#))).
 - The Diabetes UK patient resource [Alcohol and diabetes](#) ([https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/what-to-drink-with-diabetes/alcohol-and-diabetes](#)) may be helpful.

Smoking and drug misuse

- **Provide information on the impact of smoking on diabetes and its complications, such as cardiovascular disease.**
 - Advise on smoking cessation, if appropriate. See the CKS topic on [Smoking cessation](#) ([/topics/smoking-cessation/](#)) for more information. Reinforce this advice annually for people who currently do not plan to stop smoking, and at all review appointments if there is a prospect of the person stopping.
 - Advise young adult non-smokers never to start smoking.

- Provide information on the risks of drug misuse and the possible effects on blood glucose control.
 - The Diabetes UK patient resource [Recreational drugs and diabetes](https://www.diabetes.org.uk/guide-to-diabetes/young-adults/drugs) (<https://www.diabetes.org.uk/guide-to-diabetes/young-adults/drugs>) may be helpful.

Periodontitis

- Advise adults with type 2 diabetes at their annual review:
 - That they are at higher risk of periodontitis.
 - That if they get periodontitis, managing it can improve their blood glucose control and can reduce their risk of hyperglycaemia.
 - To have regular oral health reviews (their oral healthcare or dental team will tell them how often this is required).
- For adults with type 2 diabetes who have been diagnosed with periodontitis by an oral healthcare or dental team, they should be advised to attend appointments to manage and treat their periodontitis (at a frequency based on their oral health needs).

Basis for recommendation

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How should I manage an adult with a suspected hyperglycaemic emergency?

Be aware that [diabetic ketoacidosis \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/) (DKA) is more common in people with type 1 diabetes, but can occur in people with type 2 diabetes, and may present with a normal or near normal blood glucose level. [Hyperosmolar hyperglycaemic state \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/) (HHS) typically presents without significant blood or urinary ketones on testing.

- Arrange emergency admission for confirmation of the diagnosis and emergency treatment.
- After the person has recovered from an episode of DKA or HHS:
 - Discuss the [precipitating factors \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/#precipitating-factors\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/#precipitating-factors) that may have led to the episode. Consider the possibility of non-adherence to treatment, particularly if there is a history of recurrent DKA.
 - Provide information and advice about how to manage intercurrent illness and 'sick day rules', to reduce the risk of future episodes. See the section on [Managing intercurrent illness \(/topics/diabetes-type-2/management/management-adults/#managing-intercurrent-illness\)](/topics/diabetes-type-2/management/management-adults/#managing-intercurrent-illness) for more information.

Basis for recommendation

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How should I manage intercurrent illness in an adult w diabetes?

If an adult with type 2 diabetes is unwell, consider the need to arrange hospital admission or seek specialist advice, depending on clinical judgement, taking into account the person's age, co-morbidities, risk of complications, and the presence of hyperglycaemia and/or ketosis.

- **Arrange immediate hospital admission if:**
 - There is an immediate risk of [diabetic ketoacidosis \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/) (DKA), such as moderate ketonuria (2+ on urine dipstick) or ketonaemia (1.5–2.9 mmol/L) with or without hyperglycaemia, and the person cannot eat or drink.
 - There is an immediate risk of [hyperosmolar hyperglycaemic state \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/) (HHS).
 - There is suspected acute kidney injury (AKI) that cannot be managed in primary care. See the CKS topic on [Acute kidney injury \(/topics/acute-kidney-injury/\)](/topics/acute-kidney-injury/) for more information.
 - A person treated with insulin does not show signs of clinical improvement with insulin treatment. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](/topics/insulin-therapy-in-type-2-diabetes/).
- **Consider arranging hospital admission or seeking urgent specialist advice if:**
 - The underlying condition is unclear.
 - The person is dehydrated or at risk of dehydration.
 - Vomiting persists beyond 2 hours.
 - The person and their family/carers are unable to keep the blood glucose level above 3.5 mmol/L.
 - The person and their family/carers are exhausted, for example due to repeated night-time waking.
- **If admission is not needed and the person can be managed in primary care:**
 - Ensure that the person has written contact details of their specialist diabetes team, where appropriate.
 - Assess the person and manage any intercurrent illness(es), as appropriate.
 - Advise on the need to temporarily stop some drug treatments during periods of acute illness. See the section on ['Sick-day rules' \(/topics/diabetes-type-2/management/management-adults/#sick-day-rules\)](/topics/diabetes-type-2/management/management-adults/#sick-day-rules) for detailed information.
 - Provide the person with clear, individualized advice on ['sick-day rules' \(/topics/diabetes-type-2/management/management-adults/#sick-day-rules\)](/topics/diabetes-type-2/management/management-adults/#sick-day-rules) to manage diabetes during episodes of intercurrent illness or hyperglycaemia, and reinforce the advice regularly.
 - Advise that intercurrent illness may affect blood glucose control, and there is a risk for worsening hyperglycaemia even if dietary intake is reduced.
 - Ensure the person has sick day foods and drinks supplies readily accessible at home, including:
 - Easily digestible foods and sugary drinks (to provide energy and to prevent further ketosis).
 - Over-the-counter oral rehydration therapy (ORT) sachets (to prevent dehydration).
 - Glucose tablets or oral gel (to prevent hypoglycaemia).
 - Equipment for self-monitoring of blood glucose and ketones (if appropriate).
 - Additional supplies of insulin (if appropriate).
 - A glucagon kit (if appropriate).

- Note: see the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for information on blood glucose and ketone monitoring meters, glucose oral gels, and other accessories.

'Sick-day rules'

- **Provide 'sick-day rules' advice on managing episodes of intercurrent illness.**
 - Advise to temporarily stop some drug treatments during acute illness. Medication may be restarted once the person is feeling better and eating and drinking for 24–48 hours, unless there is concern about renal function.
 - On angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists (AIIAs), diuretics, or nonsteroidal anti-inflammatory drugs (NSAIDs) — stop treatment if there is a risk of dehydration, to reduce the risk of acute kidney injury (AKI).
 - On [metformin \(/topics/diabetes-type-2/prescribing-information/metformin/\)](#) — stop treatment if there is a risk of dehydration, to reduce the risk of lactic acidosis.
 - On [sulfonylureas \(/topics/diabetes-type-2/prescribing-information/sulfonylureas/\)](#) — may increase the risk of hypoglycaemia, particularly if dietary intake is reduced.
 - On [SGLT-2 inhibitors \(/topics/diabetes-type-2/prescribing-information/sglt-2-inhibitors/\)](#) — check for [ketones \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](#) and stop treatment if acutely unwell and/or at risk of dehydration, due to the risk of euglycaemic DKA.
 - On [GLP-1 receptor agonists \(/topics/diabetes-type-2/prescribing-information/glp-1-receptor-agonists/\)](#) — stop treatment if there is a risk of dehydration, to reduce the risk of AKI.
 - If on insulin therapy, do not stop treatment. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for more information.
 - Advise the dose of insulin may need to be altered during periods of illness. Seek advice from the specialist diabetes team if there is uncertainty on how to adjust insulin doses.
 - If [self-monitoring \(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](#) of blood glucose levels is indicated (for example on insulin therapy), advise:
 - An increase in monitoring frequency may be needed, such as at least every 3–4 hours including through the night and advice to record the results.
 - Insulin doses may need to be adjusted, depending on the results.
 - To continue to self-monitor blood glucose levels carefully when feeling better, until they are back to baseline.
 - Seek urgent medical advice if blood glucose levels remain uncontrolled.
 - Consider the need for blood or urinary [ketone monitoring \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](#).
 - This should be checked regularly, for example at least every 3–4 hours including through the night, and record the results.
 - If the urine ketone level is greater than 2+, or blood ketone level is greater than 3 mmol/L, the person should seek immediate medical advice. See the section on [When to suspect hyperglycaemic emergencies \(DKA and HHS\) \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/\)](#) for more information.
 - Advise to maintain their normal meal pattern (including fluids and carbohydrate intake) where possible if appetite is reduced.

- If unable to eat or vomiting, advise to replace normal meals with carbohydrate-containing drinks (such as milk, milkshakes, fruit juices, and sugary drinks).
- If blood glucose levels are high, maintain fluid intake with sugar-free fluids.
- If blood glucose levels are low, encourage regular intake of sugary fluids.
- Advise to seek urgent medical advice if the person:
 - Is unable to eat or drink, is dehydrated or at risk of dehydration.
 - Has persistent vomiting.
 - Has [hypoglycaemia \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/) that cannot be managed in primary care.
- The Diabetes UK information [Diabetes when you're unwell \(https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/illness\)](https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/illness) may be helpful.

[[Down, 2020 \(/topics/diabetes-type-2/references/\)](/topics/diabetes-type-2/references/)]

Basis for recommendation

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How should I screen for and manage complications in a diabetes?

Retinopathy

Ensure that all adults with type 2 diabetes attend for diabetic eye screening (at the time of diagnosis and at an appropriate time interval thereafter) as part of the NHS retinopathy screening programme.

- **Diabetic eye screening will be offered at diagnosis and:**
 - Every 2 years for people at low risk of sight loss (no identified diabetic retinopathy on two successive screening tests).
 - At least annually for all other people with diabetes.
- **Explain that retinal screening aims to reduce the risk of sight loss** by the early detection and treatment, if needed, of sight-threatening retinopathy.
 - The Diabetes UK patient resource [Diabetic eye screening \(https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/diabetic-eye-screening\)](https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/diabetic-eye-screening) may be helpful.
- **Arrange emergency assessment by ophthalmology if there is:**
 - Sudden loss of vision.
 - Rubeosis iridis (formation of abnormal blood vessels on the anterior iris).
 - Pre-retinal or vitreous haemorrhage.
 - Suspected retinal detachment. See the CKS topic on [Retinal detachment \(/topics/retinal-detachment/\)](/topics/retinal-detachment/) for more information.
- **Arrange an urgent referral to an ophthalmologist if there is a large sudden, unexplained reduction in visual acuity.**

Foot problems

Ensure that people with type 2 diabetes have a foot check in primary care at diagnosis and at least once a year thereafter, or sooner if any foot problems arise.

- **Give information on the risk of developing a diabetic foot problem** at diagnosis, during reviews, and if foot problems arise, including:
 - Basic foot care advice and the importance of foot care, including the need to check daily the entire surface of both feet, including areas between the toes. The Diabetes UK patient resource [Diabetes and foot problems \(https://www.diabetes.org.uk/guide-to-diabetes/complications/feet\)](https://www.diabetes.org.uk/guide-to-diabetes/complications/feet) may be helpful.
 - Foot emergencies and who to contact.
 - Footwear advice, including to avoid shoes that are too tight, have rough edges, or uneven seams; to avoid tight-fitting socks; and to change socks daily.
 - Nail-cutting advice.
 - The person's current individual risk of developing a foot problem, and how to recognize foot ulcers and pre-ulcerative signs.
- **Ask about risk factors for developing foot complications:**
 - Previous foot ulcer or lower limb amputation, peripheral arterial disease, and end-stage renal disease (ESRD). See the CKS topics on [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](/topics/peripheral-arterial-disease/) and [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information.
 - Symptoms of [peripheral neuropathy \(/topics/diabetes-type-2/background-information/complications/\)](/topics/diabetes-type-2/background-information/complications/).
 - Symptoms of peripheral arterial disease (claudication or rest pain).
 - Foot care, including personal care and nail cutting.
 - Smoking history.
- **Examine both feet — remove their shoes, socks, bandages and dressings, and examine both feet for evidence of the following risk factors:**
 - Neuropathy — use a 10 g monofilament as part of a foot sensory examination.
 - Limb ischaemia — use ankle brachial pressure index (ABPI) and interpret results carefully because calcified arteries may falsely elevate results. See the CKS topic on [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](/topics/peripheral-arterial-disease/) for more information.
 - Ulceration — may be neuropathic (most commonly affecting plantar surface or areas overlying a bony deformity); neuroischaemic or ischaemic (most commonly affecting the tips of toes or lateral border of the foot).
 - Callus.
 - Infection and/or inflammation (redness, warmth, induration, and pain/tenderness) or purulent discharge. See the CKS topics on [Paronychia - acute \(/topics/paronychia-acute/\)](/topics/paronychia-acute/) and [Cellulitis - acute \(/topics/cellulitis-acute/\)](/topics/cellulitis-acute/) for more information.
 - Note: typical findings may be reduced due to the presence of neuropathy and/or ischaemia.
 - Deformity (such as claw or hammer toes), large bony prominences, limited joint mobility.
 - Gangrene.
 - Charcot arthropathy — an acute, localized inflammatory condition that may lead to varying degrees and patterns of bone destruction, subluxation, dislocation, and deformity.
 - Footwear — ill-fitting, inadequate, or lack of.
 - If there is evidence of foot deformity and/or abnormal loading of the foot (for example callus formation), consider referral to podiatry for footwear advice and possible orthoses, specialist footwear, or splints.
 - Foot hygiene — improperly cut toenails, thickened or ingrown nails, unwashed feet, superficial fungal infection, and unclean socks. See the CKS topics on [Fungal skin](#)

[infection - foot \(/topics/fungal-skin-infection-foot/\)](/topics/fungal-skin-infection-foot/) and [Fungal nail infection \(/topics/fungal-nail-infection/\)](/topics/fungal-nail-infection/) for more information.

- **Assess the person's current risk of developing a diabetic foot problem or needing an amputation using the following risk stratification:**
 - Low risk:
 - No risk factors present except callus alone.
 - Moderate risk:
 - Deformity, *or*
 - Neuropathy, *or*
 - Peripheral arterial disease.
 - High risk:
 - Previous ulceration, *or*
 - Previous amputation, *or*
 - On renal replacement therapy, *or*
 - Neuropathy and peripheral arterial disease together, *or*
 - Neuropathy in combination with callus and/or deformity, *or*
 - Peripheral arterial disease in combination with callus and/or deformity.
 - Active foot problem:
 - Ulceration, *or*
 - Infection, *or*
 - Chronic limb-threatening ischaemia, *or*
 - Gangrene, *or*
 - Suspicion of an acute Charcot arthropathy, or an unexplained hot, swollen foot with a change in colour, with or without pain.
- **Manage the person's risk of developing a diabetic foot problem:**
 - For people at low risk:
 - Continue to carry out annual foot assessments at their annual diabetes review.
 - Emphasize the importance of good foot care.
 - Advise them that they could progress to moderate or high risk.
 - For people at moderate risk — arrange referral to the foot protection service to be seen within 6–8 weeks.
 - For people at high risk — arrange referral to the foot protection service to be seen within 2–4 weeks.
 - For people with a limb-threatening or life-threatening diabetic foot problem, refer immediately to acute services, and inform the multidisciplinary foot care service (according to local protocols and pathways) so that they can be assessed and an individualized treatment plan put in place. Examples of limb-threatening and life-threatening diabetic foot problems include:
 - Ulceration with fever or any signs of sepsis.
 - Ulceration with limb ischaemia.
 - Clinical concern that there is a deep-seated soft tissue or bone infection (with or without ulceration).
 - Gangrene (with or without ulceration).
 - For all other active diabetic foot problems, refer the person within 1 working day to the multidisciplinary foot care service or foot protection service (according to local protocols and pathways) for triage within 1 further working day.
- **Depending on the person's risk of developing a diabetic foot problem, carry out reassessments at the following intervals:**
 - Annually — for people who are at low risk, as part of their annual diabetes review.

- Frequently (for example every 3–6 months) — for people who are at moderate risk.
- More frequently (for example every 1–2 months) — for people who are at high risk, if there is no immediate concern.
- Very frequently (for example every 1–2 weeks) — for people who are at high risk, if there is immediate concern.
- Consider more frequent reassessments for people who are at moderate or high risk, and for people who are unable to check their own feet.

Diabetic kidney disease

- **Ensure all adults with type 2 diabetes have annual screening for diabetic kidney disease.**
 - See the sections on [Diagnosis \(/topics/chronic-kidney-disease/diagnosis/diagnosis/\)](/topics/chronic-kidney-disease/diagnosis/diagnosis/), [Assessment \(/topics/chronic-kidney-disease/diagnosis/assessment/\)](/topics/chronic-kidney-disease/diagnosis/assessment/), and [Management \(/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#management-in-primary-care\)](/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#management-in-primary-care) in the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information.

Cardiovascular risk factors

- **Ensure all people with type 2 diabetes have their cardiovascular risk factors assessed at least annually, including:**
 - Smoking status — offer advice on smoking cessation, if needed. See the CKS topic on [Smoking cessation \(/topics/smoking-cessation/\)](/topics/smoking-cessation/) for more information.
 - Blood glucose control. See the section on [Treatment targets \(/topics/diabetes-type-2/management/management-adults/#treatment-targets\)](/topics/diabetes-type-2/management/management-adults/#treatment-targets) for more information.
 - Blood pressure. See the CKS topic on [Hypertension \(/topics/hypertension/\)](/topics/hypertension/) for more information on diagnosing hypertension, and the section on [Antihypertensive treatment \(/topics/diabetes-type-2/management/management-adults/#antihypertensive-treatment\)](/topics/diabetes-type-2/management/management-adults/#antihypertensive-treatment) for more information on management.
 - Albuminuria. See the section on [Diabetic kidney disease \(/topics/diabetes-type-2/management/management-adults/#diabetic-kidney-disease\)](/topics/diabetes-type-2/management/management-adults/#diabetic-kidney-disease) for more information.
 - Full lipid profile (total cholesterol, high-density lipoprotein [HDL] cholesterol, low-density lipoprotein [LDL] cholesterol, and triglycerides). See the section on [Lipid modification \(/topics/diabetes-type-2/management/management-adults/#lipid-modification\)](/topics/diabetes-type-2/management/management-adults/#lipid-modification) for more information.
 - Family history of cardiovascular disease (CVD).
 - Measure height, weight, and waist circumference, and calculate the body mass index (BMI). Offer management for people who are overweight or obese. See the CKS topic on [Obesity \(/topics/obesity/\)](/topics/obesity/) for more information.
- **Ensure people with CVD risks factors are managed appropriately.** See the section on [Lifestyle advice \(/topics/diabetes-type-2/management/management-adults/#lifestyle-advice\)](/topics/diabetes-type-2/management/management-adults/#lifestyle-advice) and the CKS topic on [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](/topics/cvd-risk-assessment-management/) for more information.
 - The Diabetes UK patient resource [Diabetes and heart disease \(https://www.diabetes.org.uk/guide-to-diabetes/complications/cardiovascular_disease\)](https://www.diabetes.org.uk/guide-to-diabetes/complications/cardiovascular_disease) may be helpful.

Peripheral and autonomic neuropathy

- **Screen for symptoms of peripheral neuropathy by asking about:**

- Painful neuropathy symptoms of numbness, burning or shooting pain, tingling and/or paraesthesia of the hands and/or feet, typically in a stocking and glove distribution, often at night, which may be relieved by activity.
 - Painful neuropathy of the foot may initially affect the toes and outer edges of the feet.
 - If suspected, see the CKS topic on [Neuropathic pain - drug treatment](/topics/neuropathic-pain-drug-treatment/) (</topics/neuropathic-pain-drug-treatment/>) for more information on management options.
- There may be associated motor neuropathy, such as muscle weakness, wasting, cramps and/or twitching, which may cause eventual deformity.
 - If there is hand involvement, it may affect fine motor skills over time.
 - If there is foot involvement, consider referral to podiatry for footwear advice and possible orthoses, specialist footwear, or splints. See the section on [Foot problems](/topics/diabetes-type-2/management/management-adults/#foot-problems) (</topics/diabetes-type-2/management/management-adults/#foot-problems>) for more information on screening for peripheral neuropathy of the feet.
- The Diabetes UK patient resource [Diabetic neuropathy](https://www.diabetes.org.uk/guide-to-diabetes/complications/nerves_neuropathy) (https://www.diabetes.org.uk/guide-to-diabetes/complications/nerves_neuropathy) may be helpful.
- **Assess for symptoms or signs suggesting autonomic neuropathy**, which can affect multiple systems and typically has a gradual, progressive onset:
 - Postural hypotension (may be defined as a drop in systolic blood pressure of more than 30 mmHg when changing from a lying to a standing position, without any increase in heart rate).
 - Ask about drug treatment with tricyclic antidepressants or antihypertensive drugs that may contribute to symptoms.
 - Consider arranging referral to the specialist diabetes team if symptoms are troublesome and affecting daily functioning.
 - 'Diabetic gastroparesis' suggested by delayed gastric emptying, bloating, nausea, and post-prandial vomiting.
 - Advise a small-particle-size diet (mashed or pureed food) for symptom relief of delayed gastric emptying. Consider arranging referral to a gastroenterologist or the specialist diabetes team for consideration of treatment with a prokinetic drug.
 - Lower gastrointestinal involvement may cause lower abdominal pain, unexplained diarrhoea (may be nocturnal), and faecal incontinence.
 - Unexplained urinary symptoms such as hesitancy, reduced frequency, inadequate bladder emptying, and urinary retention, if other possible causes have been excluded. See the CKS topics on [LUTS in men](/topics/luts-in-men/) (</topics/luts-in-men/>), [Urinary tract infection \(lower\) - men](/topics/urinary-tract-infection-lower-men/) (</topics/urinary-tract-infection-lower-men/>), and [Urinary tract infection \(lower\) - women](/topics/urinary-tract-infection-lower-women/) (</topics/urinary-tract-infection-lower-women/>) for more information.
 - If suspected, consider arranging referral to the specialist diabetes team or urology for advice on management.
 - Sexual dysfunction, erectile dysfunction, and/or retrograde ejaculation.
 - See the CKS topic on [Erectile dysfunction](/topics/erectile-dysfunction/) (</topics/erectile-dysfunction/>) for more information on diagnosis and management.
 - The Diabetes UK patient resources [Diabetes and sexual problems - in women](https://www.diabetes.org.uk/guide-to-diabetes/complications/sexual-problems-women) (<https://www.diabetes.org.uk/guide-to-diabetes/complications/sexual-problems-women>) and [Diabetes and sexual problems - in men](https://www.diabetes.org.uk/guide-to-diabetes/complications/sexual-problems-men) (<https://www.diabetes.org.uk/guide-to-diabetes/complications/sexual-problems-men>) may be helpful.
 - Sweating abnormalities, such as reduced sweating in a stocking and glove distribution.

- Impaired awareness of [hypoglycaemia \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/\)](/topics/diabetes-type-2/diagnosis/when-to-suspect-hypoglycaemia/).
 - If suspected, arrange referral to the specialist diabetes team for advice on management.
 - The Diabetes UK patient resource [What is a hypo? \(https://www.diabetes.org.uk/guide-to-diabetes/complications/hypos\)](https://www.diabetes.org.uk/guide-to-diabetes/complications/hypos) may be helpful.

Basis for recommendation

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How should I manage cardiovascular risk in an adult with diabetes?

Antihypertensive treatment

If the person is already on antihypertensive drug treatment:

- Reinforce the importance of [lifestyle advice \(/topics/diabetes-type-2/management/management-adults/#lifestyle-advice\)](/topics/diabetes-type-2/management/management-adults/#lifestyle-advice).
- Review blood pressure control and drug treatments used. See the CKS topic on [Hypertension \(/topics/hypertension/\)](/topics/hypertension/) for more information on monitoring blood pressure and drug treatment options.
 - Amend treatment if there is uncontrolled blood pressure and/or drug treatment is not appropriate due to microvascular or metabolic [complications \(/topics/diabetes-type-2/background-information/complications/\)](/topics/diabetes-type-2/background-information/complications/).
- Aim to maintain blood pressure below the following targets, depending on the person's age, risks and benefits of treatment, and co-morbidities (including frailty and multimorbidity):
 - For adults with a urine albumin:creatinine ratio (ACR) less than 70 mg/mmol, aim for a clinic systolic blood pressure less than 140 mmHg (target range 120 to 139 mmHg) and a clinic diastolic blood pressure less than 90 mmHg.
 - For adults with an ACR of 70 mg/mmol or more, aim for a clinic systolic blood pressure less than 130 mmHg (target range 120 to 129 mmHg) and a clinic diastolic blood pressure less than 80 mmHg.
 - In adults aged 80 or more, whatever the ACR, aim for a clinic systolic blood pressure less than 150 mmHg (target range 140 to 149 mmHg) and a clinic diastolic blood pressure less than 90 mmHg.
- The Diabetes UK patient resource [Diabetes and blood pressure \(https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/blood-pressure\)](https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/blood-pressure) may be helpful.

If the person has newly diagnosed hypertension:

- Reinforce the importance of [lifestyle advice \(/topics/diabetes-type-2/management/management-adults/#lifestyle-advice\)](/topics/diabetes-type-2/management/management-adults/#lifestyle-advice).
- Confirm a diagnosis of hypertension in a person with clinic blood pressure of 140/90 mmHg or higher *and* ABPM daytime average or HBPM average of 135/85 mmHg or higher.
 - Measure standing as well as seated blood pressure.
 - Classify the person's stage of hypertension.

- Assess whether antihypertensive treatment is needed, depending on the stage of hypertension, person's age, estimated 10-year cardiovascular risk, evidence of target organ damage, and co-morbidities (including frailty or multimorbidity).
- See the CKS topic on [Hypertension \(/topics/hypertension/\)](/topics/hypertension/) for more information on stages of hypertension.
- If antihypertensive drug treatment is indicated:
 - Assess the person's treatment goals and preferences and explain the risks and benefits of drug treatment. See the CKS topic on [Hypertension \(/topics/hypertension/\)](/topics/hypertension/) for more information on drug treatment options.
 - Offer a generic angiotensin-converting enzyme (ACE) inhibitor or angiotensin-II receptor antagonist (AIIA) to all adults (of any age or ethnic origin).
 - For adults of black African or African-Caribbean origin, consider an AIIA in preference to an ACE inhibitor.
 - If an ACE inhibitor is not tolerated, offer an AIIA.
 - Do not combine an ACE inhibitor with an AIIA to treat hypertension.
 - Offer a calcium-channel blocker (CCB) or thiazide-like diuretic in addition to an ACE inhibitor or AIIA, if hypertension is not controlled by monotherapy.
 - Offer a combination of an ACE inhibitor or AIIA, a CCB *and* a thiazide-like diuretic if hypertension remains uncontrolled.
 - If hypertension is not controlled with optimal tolerated doses of an ACE inhibitor or an AIIA, plus a CCB and a thiazide-like diuretic:
 - Before considering further treatment, confirm elevated clinic blood pressure measurements using ABPM or HBPM, assess for postural hypotension, and discuss adherence with treatment.
 - Consider adding a fourth antihypertensive drug, or seek specialist advice, for people with confirmed resistant hypertension. See the CKS topic on [Hypertension \(/topics/hypertension/\)](/topics/hypertension/) for more information.
 - Seek specialist advice if blood pressure remains uncontrolled with optimal tolerated doses of four antihypertensive drugs.
 - The Diabetes UK patient resource [Diabetes and blood pressure \(https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/blood-pressure\)](https://www.diabetes.org.uk/guide-to-diabetes/managing-your-diabetes/blood-pressure) may be helpful.

Lipid modification

- If an adult with type 2 diabetes does *not* have established cardiovascular disease (CVD), offer atorvastatin 20 mg once daily for primary prevention of CVD if:
 - The person is aged 84 years and younger, and their estimated 10-year risk of developing CVD is 10% or more.
 - See the CKS topic on [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](/topics/cvd-risk-assessment-management/) for more information on assessing CVD risk and using the QRISK assessment tool.
 - The person is 85 years of age or older, taking into account the person's preferences, benefits and risks of treatment, and co-morbidities (including frailty and multimorbidity).
 - The person has a diagnosis of chronic kidney disease (CKD).
 - See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information.
 - Note: see the CKS topic on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for detailed information on baseline blood tests, initiating statin treatment, and monitoring.

- If an adult with type 2 diabetes has established CVD, offer atorvastatin 80 mg once daily for secondary prevention of CVD.
 - See the CKS topics on [Angina \(/topics/angina/\)](/topics/angina/), [Stroke and TIA \(/topics/stroke-tia/\)](/topics/stroke-tia/), and [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](/topics/peripheral-arterial-disease/) for more information.
 - Note: see the CKS topic on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for detailed information on baseline blood tests, initiating statin treatment, and monitoring.
- Aim for a greater than 40% reduction in non-HDL cholesterol.
 - Consider up-titration of the atorvastatin dose to achieve this target, if appropriate, depending on clinical judgement. See the CKS topic on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for more information.
- The Diabetes UK patient resource [Cholesterol and diabetes \(https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/eating-with-diabetes/managing-other-medical-conditions/cholesterol-and-diabetes\)](https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/eating-with-diabetes/managing-other-medical-conditions/cholesterol-and-diabetes) may be helpful.

Antiplatelet treatment

- Do not routinely offer antiplatelet treatment (aspirin or clopidogrel) for the primary prevention of cardiovascular disease (CVD) in adults with type 2 diabetes.
 - See the CKS topic on [Antiplatelet treatment \(/topics/antiplatelet-treatment/\)](/topics/antiplatelet-treatment/) for more information on the primary and secondary prevention of CVD in adults with type 2 diabetes.

Basis for recommendation

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